VF Asia Footwear Laboratory Testing Manual

Test Method for Peel Strength of Outsole to Upper and to Midsole	Test Method #: VF- F02 Page 1 of 4
Date Issued : May,2019 Revision # :2 Reason for Revision: Rectify the unit of measure from kg to kgf for strength of material.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the level of adhesion that a material (typically an upper or midsole material) can achieve when bonded to a standard rubber material, which represents the outsole.

Test samples are prepared using controlled conditions and techniques and are evaluated using the 180° peel strength (T-peel) technique.

REFERENCES: DIN 53273&53578 / SATRA AM1

EQUIPMENT AND MATERIAL:

1. Coating device for applying the primers and adhesives to specific uniform thicknesses, if available, a metered coating bar should be used, If a brush is to be used, care must be taken to ensure that a uniform, continuous application of adhesive is used.
2. Air-circulating oven, capable of maintaining a temperature of ±1°C.
3. Thermocouple/surface pyrometer, capable of measuring the surface temperature of a part to ±1°C.
4. Press capable of applying a pressure of 2-4kgf/cm^2 over the entire bond surface, that each cm^2 of the bond sample must have a load of 2-4 kgf/cm^2 applied to it. For example, a test sample which is 10 cm* 10cm, requires a load of 200-400 kgf.
5. Clean, lint-free cloth
6. Cleaning solvents, toluene free.
7. Wire wheel buffing machine or sanding belt.
8. Balance, capable of weighing to 0.01 grams.
9. Clean, dry stirring rod.
10. Tensile testing machine, meeting the requirements of ASTM D 638
11. Clamping jaws for tensile tester
12. Steel scale, capable of measuring to 0.5mm.
13. Upper, outsole, midsole

SAMPLE PREPARATION:

1. Condition the sample for at least 48 hours at the standard laboratory condition at 23.0±2.0˚C Temperature and 55 ±10 % Relative Humidity.
2. Test substrates should be die cut or hand cut 150 ± 50 mm in length and 25 ± 5 mm width.
3. Surfaces of test substrates should be cleaned and buffed. If information concerning a specific material or combination of materials is not available with VF, prepare the surfaces according to the directions of the adhesive/primer supplier.)

Test Method for Peel Strength of Outsole to Upper and to Midsole	Test Method #: VF- F02 Page 2 of 4
Date Issued : May,2019 **Revision # :2** **Reason for Revision: Rectify the unit of measure from kg to kgf for strength of material.**	**Created by: Nila Babia Erazo**

4. Cleaning must always be performed with a clean, lint-free cloth and uncontaminated solvent-toluene free, thorough drying of the surface is required prior to proceeding to the next step.
5. Cleaning must always be performed with a clean, lint-free cloth and uncontaminated solvent-toluene free, thorough drying of the surface is required prior to proceeding to the next step.
6. Buffing must always be performed uniformly and in the longitudinal direction of the test samples /strips, all debris, dust, and other contamination from buffing must be removed from the substrate before proceeding to the next step, Substrates must be primed within 60 minutes following buffing.
7. Priming should be performed according to the **VF Bonding Process** for material. If information concerning a specific material or combination of materials is not available, prime the surfaces according to the directions of the primer supplier). When bonding rubber, cover the entire bonding surface with a thin, continuous application of the halogenation primer, Allow the primer to dry for 35 ±5 minutes at 23 ±2°C, prior to applying the adhesive.
   Note : When halogenation priming is performed ,the primer is not to be used until 10 minutes after the addition and initial mixing of the halogenation power ,The pot life of the primer is 2 hours .Any primer which has not been used within 2 hours must be discarded.
8. Two-component adhesives should be prepared by adding 4.0 ±0.1% hardener by weight (unless otherwise specified), Following addition of the hardener, the adhesive should be thoroughly mixed by hand stirring for at least 60 seconds, Upon completion of mixing, the should then be applied uniformly and continuously to the entire bonding surface, preferably using a metered coating bar, Any adhesive which has not been used within 30 minutes of mixing must be discarded.
   Note: Because the samples will need to be inserted into the jaws of a tensile tester, the samples should be assembled such that no bond is present at one end of the test samples for a distance of 35 ±5mm, this can be most easily accomplished by not applying adhesive to this portion of the test sample.

9. Dry the adhesive at 23 ±2^{0}C for 35+/-5 minutes (or oven dry at the recommended conditions of the adhesive supplier) before heat activation of the adhesive.

10. Place the bond samples in an oven at 70 ±1^{0}C until the adhesive surface reaches 60 ±2^{0}C. The surface temperature is measured using a surface pyrometer.

11. Assembled and press the bond samples immediately (<10 seconds) upon removal from the oven. Pressing must be undertaken for 10 seconds using a pressure of 2-4kgf/cm2, That is,

Test Method for Peel Strength of Outsole to Upper and to Midsole	Test Method #: VF- F02 Page 3 of 4
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each cm2 of the bond sample must have a load of 2-4 kgf applied to it, for example, a test sample which is 10cm*10cm requires a load of 200-400 kgf.

12. Prepared bond samples must be conditioned at 23 ±2 ^{0}C and 55 ±10% humidity for a minimum of 24 hours prior to proceeding to subsequent evaluations.
13. Following conditioning, die or hand cut test samples which are 15 cm in length by 2.5-3.0 cm in width.

REPETITIONS: Prepare five (5) bond samples per material combination. If aging/Hydrolysis evaluations are to be undertaken, five un-aged (control) samples and five aged samples must be prepared per adhesive or material combination

Note: If the test material is considered directional (5) samples should be prepared in both the length and width directions.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A. Un-aged Testing

1. Measure the width of the test sample to the nearest 0.05mm at five equidistant locations along the entire length of the sample. Record the average of the five readings to the nearest 0.05mm (**W**)

2. Zero and calibrate the tensile testing machine. Set the crosshead speed to 100mm/minute, the jaw gap to approximately 3cm, and the load range to 20 kgf.

3. Centrally clamp approximately 20mm of the rubber portion (or stretchy material) of the bond sample into the stationary jaw of the tensile tester. Centrally clamp approximately 20mm of the free end of the second substrate of the bond sample into the moving jaw of the tensile tester. The sample must be inserted into the jaws such that the bond line can be observed by the operator as peeling progresses.

4. Operate the tensile testing machine at a speed of 100 mm/minute.

Test Method for Peel Strength of Outsole to Upper and to Midsole	Test Method #: VF- F02 Page 4 of 4
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5. Continue testing until the bond sample has been completely separated.

Disregard the reading over the initial 1 cm of the sample and the final 1 cm of the sample.

For each test sample, determine the average of the three (3) minimum peaks and or plateaus (**L**) in kgf.

In addition, the mode of failure should be noted, follows:

AF = Failure at the adhesive/substrate interface or the adhesive/primer interface

CF = Cohesive failure of the adhesive (failure within the adhesive or between the two layers of adhesive)

PF = Failure at the primer /substrate interface

SF = Failure of the substrate, including the location of the failure within the substrate.

If identified, the reason for bond failure should also be reported. For example, insufficient roughing (IR), insufficient activation temperature (IT) inadequate pressing (IP) under application of adhesive (UA) or primer (UP) ,etc.

CALCULATONS:

Peel Strength (kgf /cm) = L/ W

REPORT: The peel strengths in kgf /cm taken in every 2 cm